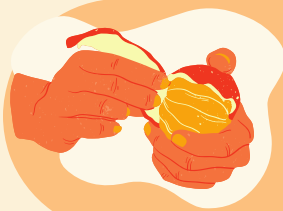


LIMONENE

THE MOLECULE BEHIND THE CITRUS SCENT

compound identity

- **Common name:** Limonene
- **IUPAC name:** 1-Methyl-4-(prop-1-en-2-yl)cyclohex-1-ene
- **Molecular formula:** $C_{10}H_{16}$
- **Molar mass:** 136.234 g/mol
- **CAS number:** 138-86-3



Structure

- **Alkene (C=C)** — limonene contains two carbon-carbon double bonds.
- **Cyclohexene ring** — one of the double bonds is within the six-membered ring.
- **Isopropenyl group** — the second double bond is found in the side chain.

Important Properties

- **Appearance:** Colorless to pale-yellow liquid
- **Odor:** Characteristic citrus/orange-like odor
- **Molecular type:** Nonpolar hydrocarbon
- **Volatile:** readily evaporates compared with less-volatile organic compounds



What is Amazing About it

- A citrus peel can become a valuable chemical resource.
- Limonene is a major component of citrus essential oils, particularly orange peel oil. Instead of treating orange peels only as waste, they can serve as a source of valuable limonene.



Uses and Value

- 🍊 Flavoring: Used to provide citrus-like flavor.
- 🌸 Fragrance: Used in perfumes and scented products because of its citrus odor.
- 🧼 Cleaning products: Used as a fragrance ingredient and solvent in some formulations



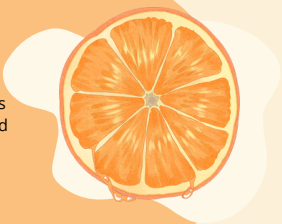
Risks and Safety



- Flammable liquid and vapor
- Can cause skin irritation in some people.
- Can cause serious eye irritation.
- Oxidized limonene may be more likely to cause skin sensitization.

Chemical Engineering Connection

- Extraction → Separation → Purification
- Limonene can be recovered from orange peels through extraction, followed by separation and purification.
- A chemical-engineering process can involve:
- Orange peel waste → Extraction → Separation → Purification → Limonene product



REFERENCE



References

- Siddiqui, S. A., Pahmeyer, M. J., Assadpour, E., & Jafari, S. M. (2022). Extraction and purification of d-limonene from orange peel wastes: Recent advances. Industrial Crops and Products, 177, 114484. <https://doi.org/10.1016/j.indcrop.2021.114484>